

CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture M2c: Conditional Probability and Bayes' Theorem

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College of Engineering

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Outline

- ① Conditional probability
- ② Independent events
- ③ Multiplication rule
- ④ Theorem of total probability
- ⑤ Bayes' theorem
- ⑥ Outlook

Recap from Lecture 2b: Theory of Probability

- Three axioms of probability:

$$P(E) \geq 0 \quad \text{and} \quad P(E) \leq 1$$

$$P(S) = 1$$

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \dots + P(E_n) \quad (\text{Mutually exclusive})$$

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 - Combinations (distinct; order not important) of group of k items chosen from set of n items: $n!/(k!(n-k)!)$

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- Understand Bayes' Theorem and learn how to apply it to solving inverse probability problems

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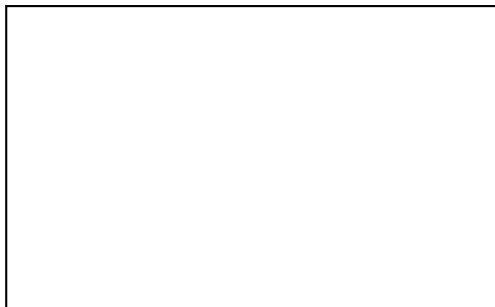
$P(E_2|E_1)$ = the conditional probability of E_2 given E_1

Conditional probability (cont.)

We can think of $P(E_1|E_2)$ as the probability of realizing sample points of E_1 within the subsample space of E_2 .

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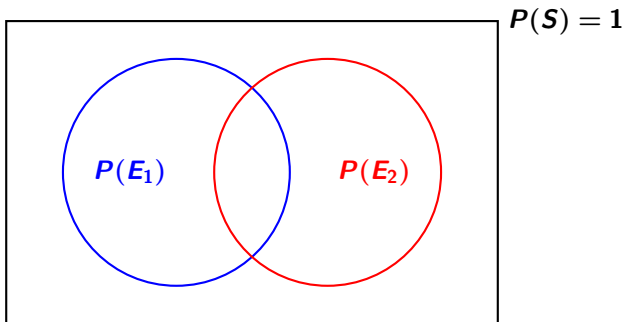
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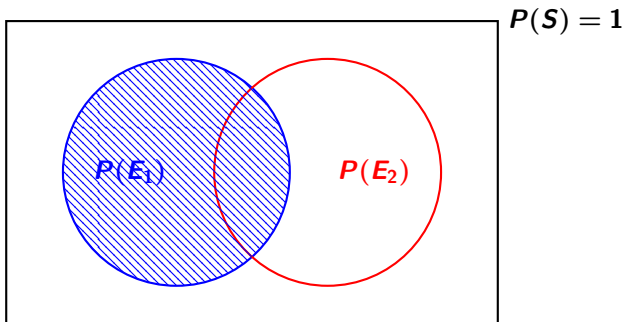
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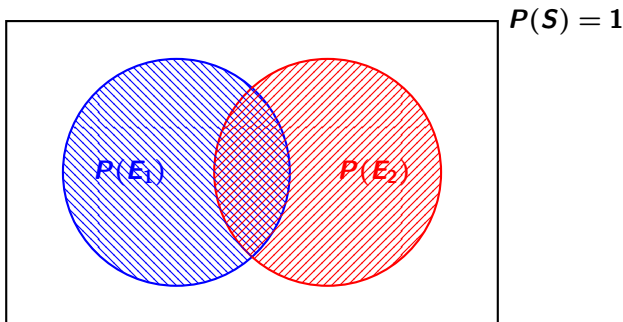
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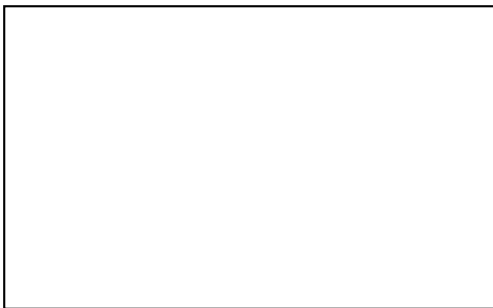
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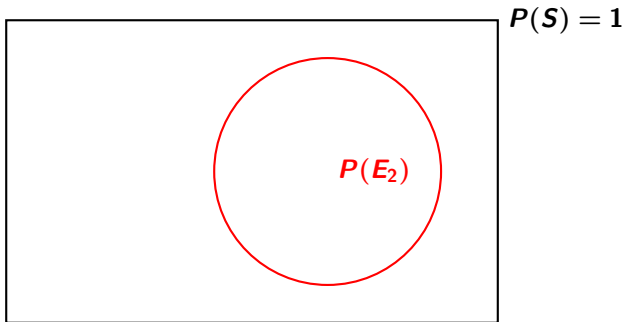


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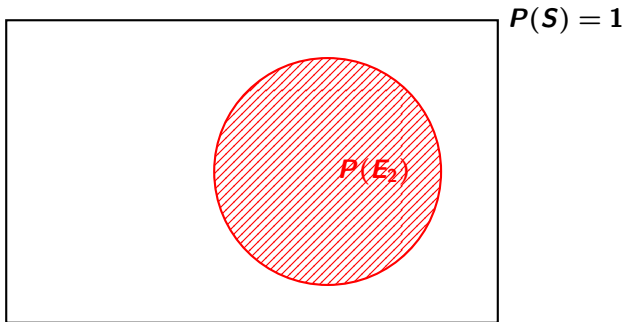
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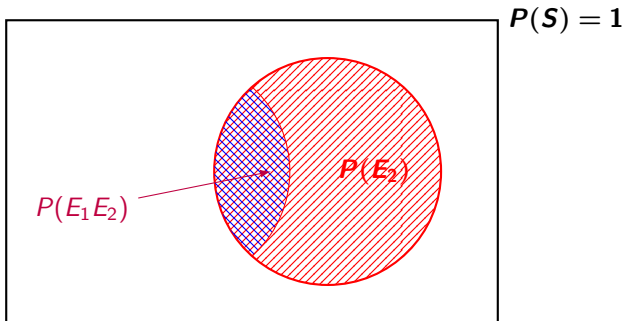
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 - The numerator is simply the intersection of both events

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There are two routes from City A to City B. One or both routes may be closed due to severe weather.

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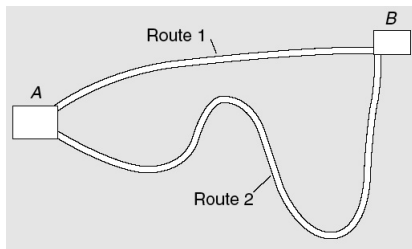
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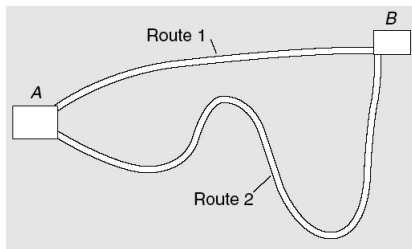
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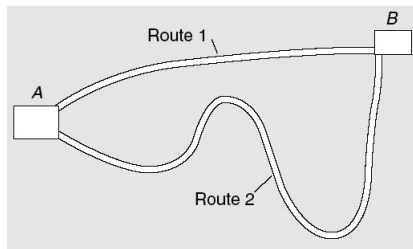


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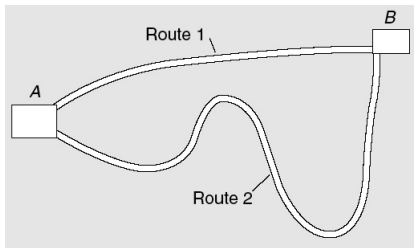
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- (a) What is probability that Route 1 is open during a storm given that Route 2 is also open?
- (b) What is probability that Route 2 is open during a storm conditioned on the event that Route 1 is also open?
- (c) What is probability that Route 1 is closed during a storm given that Route 2 is also closed?

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(c) Use the conditional probability formula to find $P(Y = 1|X = 1)$

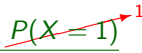
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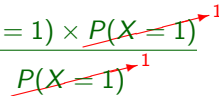
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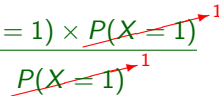
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X and Y are independent events.

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$$P(AB) = P(A)P(B) \quad (12)$$

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Equivalently:

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Statistically independent events

IF A and B are statistically independent, the multiplication rule becomes:

$$P(AB) = P(A)P(B) \quad (12)$$

That is, the **joint probability** of two statistically independent events is the **product** of their individual probabilities.

The multiplication rule: three events

The probability of the joint occurrence of three events is:

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$$\begin{aligned}P(ABC) &= P(A|BC)P(BC) \\ &= P(A|BC)P(B|C)P(C)\end{aligned}\tag{13}$$

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Statistically independent events

If the three events are statistically independent, then:

$$P(ABC) = P(A)P(B)P(C)\tag{14}$$

The multiplication rule: conditional probabilities

Equally applies to probabilities of events conditioned on the same subsample space.

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Statistically independent events

If A and B are statistically independent, the multiplication rule for two events conditioned on the same space becomes:

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Example 3: Airline industry strikes

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The airline industry in a certain country is subject to labor strikes by the pilots (event A), mechanics (event B) or flight attendants (event C).

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The probability of strikes by each of the individual groups in the next 3 years is given by:

$$P(A) = 0.03 \quad P(B) = 0.05 \quad P(C) = 0.06$$

Assuming that events A , B and C are statistically independent, find

- (a) the probability that all groups will strike in the next 3 years
- (b) the probability that none of the groups will strike in the next 3 years
- (c) the probability of a labor strike in the airline industry in the next 3 years (i.e., that any of the groups will strike)

Example 3: Airline industry strikes (cont.)

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$$P(ABC) = P(A)P(B)P(C) \quad \text{statistical independence}$$

Example 3: Airline industry strikes (cont.)

$$P(A) = 0.03 \quad P(B) = 0.05 \quad P(C) = 0.06$$

(a) The probability that all 3 groups will strike in the next 3 years is given by:

$$\begin{aligned} P(ABC) &= P(A)P(B)P(C) \quad \text{statistical independence} \\ &= 0.03(0.05)(0.06) \end{aligned}$$

Example 3: Airline industry strikes (cont.)

$$P(A) = 0.03 \quad P(B) = 0.05 \quad P(C) = 0.06$$

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Example 3: Airline industry strikes (cont.)

$$P(A) = 0.03 \quad P(B) = 0.05 \quad P(C) = 0.06$$

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(Step 1) Formulate the desired probability:

$$P(A \cup B \cup C)$$

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(c) The probability of a labor strike is the probability that any combination of the groups will strike.

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(Step 2) Realize that this is simply the probability of the complement of the event in which none of the 3 groups strike, which you have already computed.
Thus:

Example 3: Airline industry strikes (cont.)

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Thus:

$$P(A \cup B \cup C) = 1 - P(\bar{A})P(\bar{B})P(\bar{C})$$

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Example 3: Airline industry strikes (cont.)

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(c) The probability of a labor strike is the probability that any combination of the groups will strike. [ALTERNATE SOLUTION using de Morgan's rule]

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$$P(A \cup B \cup C) = 1 - P(\overline{A \cup B \cup C}) \quad \text{complement rule}$$

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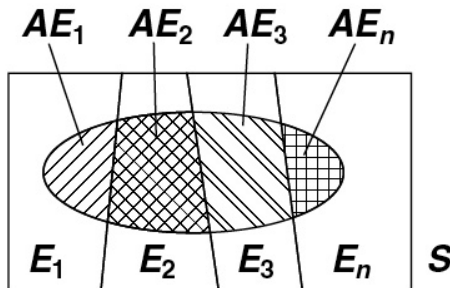
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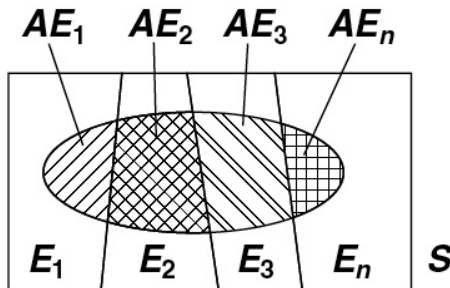
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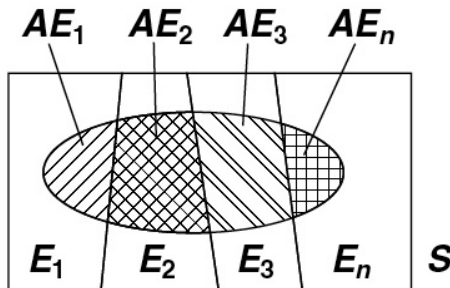
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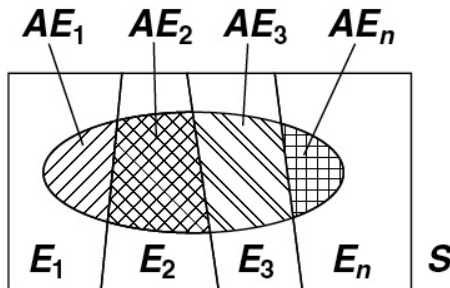
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$$P(A) = P(AE_1) + P(AE_2) + \dots + P(AEn)$$

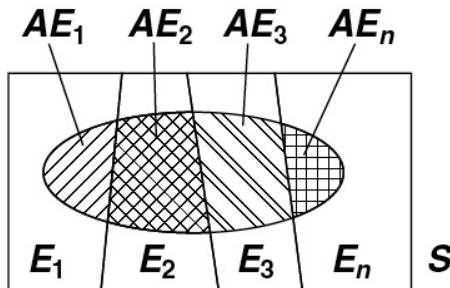
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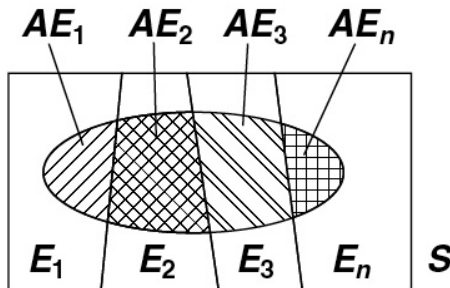
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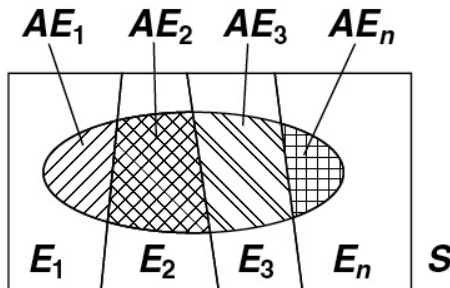
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Note that:

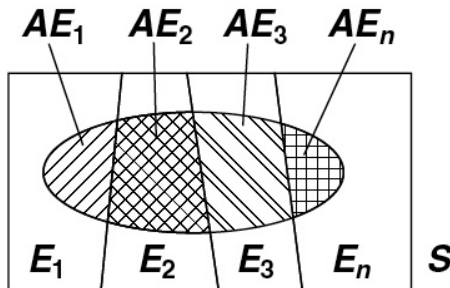
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Note that:

$$P(AE_1) = P(A|E_1)P(E_1),$$

etc.

Example 4: Flooding and snow accumulation

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$$P(F|H) = 0.90 \quad P(F|N) = 0.40 \quad P(F|L) = 0.10$$

and in the winter

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$$P(H) = 0.20 \quad P(N) = 0.50 \quad P(L) = 0.30$$

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Find the probability of flooding in the river during the spring season $P(F)$.

Example 4: Flooding and snow accumulation (cont.)

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We use the theorem of total probability:

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$$P(F) = P(F|H)P(H) + P(F|N)P(N) + P(F|L)P(L)$$

Example 4: Flooding and snow accumulation (cont.)

We use the theorem of total probability:

$$\begin{aligned} P(F) &= P(F|H)P(H) + P(F|N)P(N) + P(F|L)P(L) \\ &= 0.90(0.20) + 0.40(0.50) + 0.10(0.30) \end{aligned}$$

Example 4: Flooding and snow accumulation (cont.)

We use the theorem of total probability:

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Derivation of Bayes' theorem

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This is in essence Bayes' Theorem

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Rev. Thomas Bayes
(1701-61)

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If the event E can be conditioned on only two events A and B , then:

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$$P(B|E) = \frac{P(E|B)P(B)}{P(E|A)P(A) + P(E|B)P(B)} \quad (24)$$

Example 5: Construction supplies

Aggregates for the construction of a reinforced concrete building are supplied by two companies.

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E = aggregates are substandard

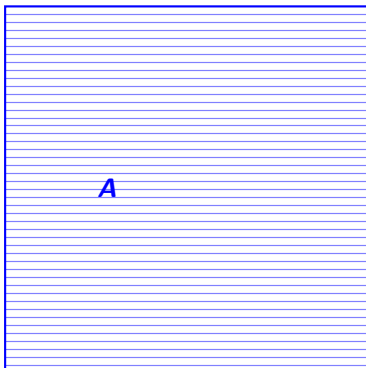
Example 5: Construction supplies (cont.)

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- (a) Draw a Venn diagram and convince yourself that
 $P(A) = 0.60$, $P(B) = 0.40$, $P(E|A) = 0.03$, $P(E|B) = 0.01$

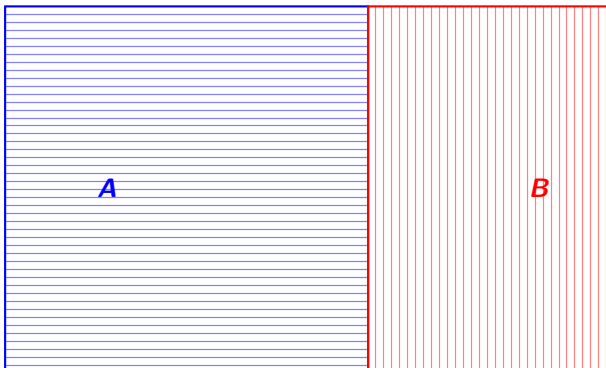
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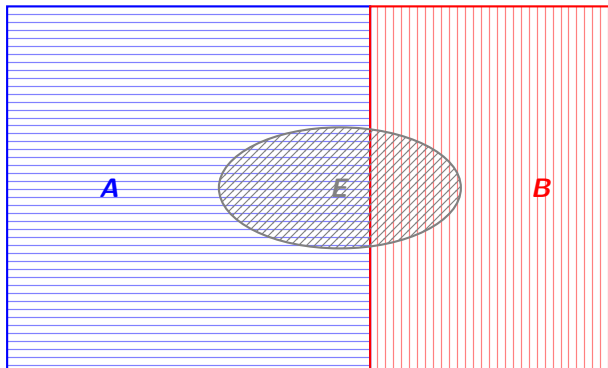
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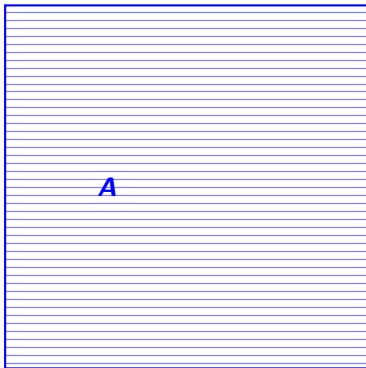
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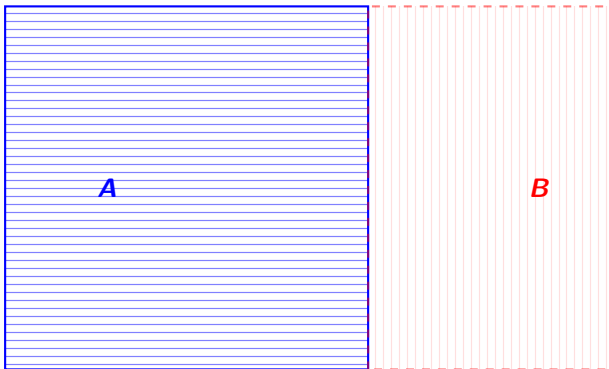
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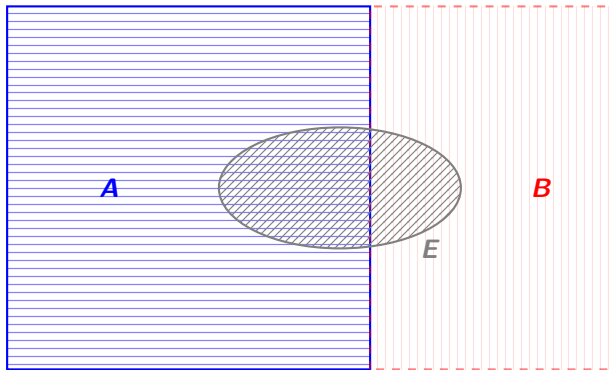
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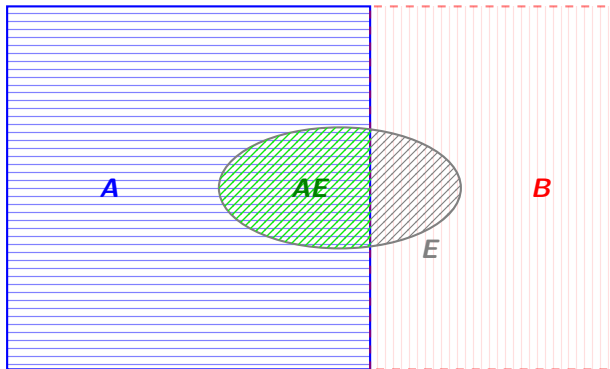
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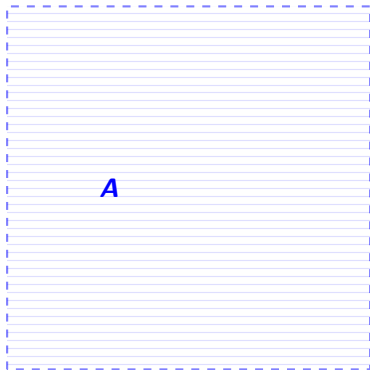
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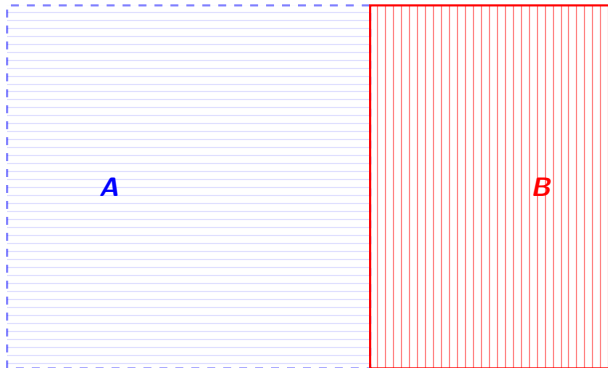
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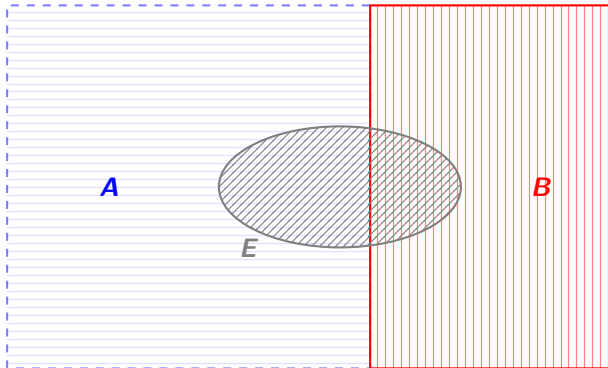
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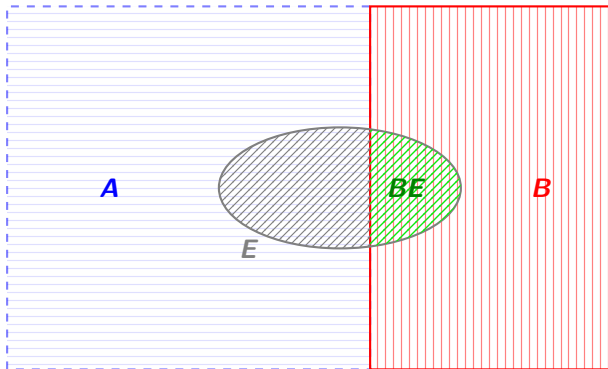
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(b) Show that $P(E) = 0.022$

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We use the Theorem of Total Probability:

$$P(E) = P(E|A)P(A) +$$

Example 5: Construction supplies (cont.)

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We use the Theorem of Total Probability:

$$P(E) = P(E|A)P(A) + P(E|B)P(B)$$

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(b) Show that $P(E) = 0.022$

We use the Theorem of Total Probability:

$$\begin{aligned} P(E) &= P(E|A)P(A) + P(E|B)P(B) \\ &= (0.03)(0.6) + \end{aligned}$$

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Example 5: Construction supplies (cont.)

(c) Show that the probability $P(A|E) = 0.82$ and discuss its significance

Here, we use Bayes' Theorem:

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$$\begin{aligned}
 P(A|E) &= \frac{P(E|A)P(A)}{P(E|A)P(A) + P(E|B)P(B)} \\
 &= \frac{P(E|A)P(A)}{P(E)} \quad \text{Denominator: total probability}
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 &= 0.818 \approx \boxed{0.82}
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 &= 0.818 \approx \boxed{0.82}
 \end{aligned}$$

$P(A|E)$ is the posterior probability of A having observed E . In other words, having prior knowledge of A (i.e. $P(A)$) and the likelihood of E given A , we can update our knowledge of A (posterior probability) based on these observations.

Recap

- **Conditional probability:**

Recap

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$$P(A|B) = \frac{P(AB)}{P(B)} \quad (25)$$

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$$P(AB) = P(A)P(B) \quad (26)$$

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$$P(A|B) = \frac{P(AB)}{P(B)} \quad (25)$$

- **Independent events:**

$$P(AB) = P(A)P(B) \quad (26)$$

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$$P(AB) = P(A|B)P(B) = P(B|A)P(A) \quad (28)$$

Recap (cont.)

- Total probability:

$$P(E) = P(E|A_1)P(A_1) + P(E|A_2)P(A_2) + \cdots + P(E|A_n)P(A_n) \quad (29)$$

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- Bayes' Theorem:

$$P(A_1|E) = \frac{P(E|A_1)P(A_1)}{P(E|A_1)P(A_1) + P(E|A_2)P(A_2) + \cdots + P(E|A_n)P(A_n)} \quad (30)$$